

Varun Sathya

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EDUCATION

College of Engineering Trivandrum

B.Tech Electrical & Electronics Engineering — GPA: 8.88/10

Relevant Coursework: Power Electronics, Linear Control Systems, Power Systems, Digital Electronics, Analog Electronics, Microprocessors and Microcontrollers, Signals & Systems

SKILLS

Tools & Software: MATLAB/Simulink, KiCad, Proteus, SPICE Simulation, Verilog (Quartus/ModelSim)

Technical Areas: PID & Feedback Control Systems, DC-DC Converters, MPPT, Heuristic Optimization (GA, PSO, Simulated Annealing)

Languages: English, Tamil, Hindi, Malayalam

EXPERIENCE

Indian Institute of Space Science and Technology

May 2026 – Present

Research Intern — Modular Electrical Power Distribution for Small Satellites

- Conducting literature survey and simulation of modular EPS architectures for small satellites (Project AV34, under Dr. Anindya Dasgupta); initiated redesign of InspireSat-3, focused on improved modularity and scalability.
- Developed simulation models for dual solar PV arrays with P&O MPPT, Li-ion battery charge/discharge dynamics, and OCV-based SOC estimation; integrated and validated three LT3652 ICs for MPPT-enabled charging from two solar panels.
- Designed power converters for InspireSat-3 payload requirements; currently developing KiCad PCB layouts for modular power distribution units.

Indian Institute of Space Science and Technology

June 2025

Summer Research Intern

- Developed agent-based models in NetLogo simulation software for [Computational Models of Complex Societies](#).
- Gained expertise in heuristic optimization algorithms such as Genetic Algorithms, Particle Swarm Optimization, and Simulated Annealing.

PROJECTS

PID-Controlled DC-DC Boost Converter with Genetic Algorithm Tuning

April 2026

Course Project — Dept. of Electrical Engineering, CET

- Designed and simulated closed-loop PID boost converter in MATLAB/Simulink (12V to regulated 20V output).
- Implemented Genetic Algorithm for optimal PID tuning to minimize steady-state error and overshoot.
- Developing 2-layer KiCad PCB and STM32 firmware in C for 25kHz PWM generation controlled by GA.

Solar PV MPPT with DC-DC Converter

Mar 2026 – Present

Project Coordinator — IET On-Campus CET

- Leading a team to develop a solar Maximum Power Point Tracking energy harvesting system
- Designing hardware modules for DC-DC boost converters, DC-AC inverters, and LCL-filters
- Designing control algorithms for Perturb and Observe MPPT in MATLAB and Python for efficient power extraction

Deep Learning for Autonomous Agricultural Drone

Jan 2025

Delta Automations — State Government Funded Project

- Contributed to development of autonomous agricultural drone for weed detection and removal.
- Built semantic segmentation pipeline using DeepLabv3 architecture.
- Trained model on large image dataset for pixel-level healthy vs weed-infected region classification.

VLSI Macro Floorplanning Tool (LAIOpt)

Dec 2025 – Jan 2026

1st Place — TCS Decode Hackathon & Winner — VIA CampusConnect

- Developed hybrid Reinforcement Learning + Simulated Annealing optimizer for automating macro placement in VLSI Floorplanning.
- Minimized wirelength and thermal dissipation.
- Deployed the tool as a Streamlit web application.

ACTIVITIES

- As a personal project, researched and benchmarked Fuzzy Logic, Genetic Algorithm, and PSO for automated PID parameter tuning for power converters in MATLAB/Simulink.
- Built embedded control systems including Arduino-based self-balancing robot (gyroscope + PID), line follower robot, obstacle avoidance robot etc. as a part of IEEE SB CET and RoboCET.
- Prototyped geofencing safety device using Arduino Nano, GPS, and GSM modules with Proteus simulation, submitted to project expo at NIT Calicut (Tathva 2024)